Project Based Learning

On

RCC BEAM DESIGN AND DETAILING USING AUTOCAD

By

Chaoutakuri Pramod - 25X01A0114

Under the guidance of

SHAIK KHALEEL

Autocad Trainer

Department of Civil Engineering

Narsimha Reddy Engineering College (UGC-Autonomous)
Approved by AICTE, Affiliated to JNTUH, Accredited by NBA &
NAAC with A-Grade

Maisammaguda (V), Kompally, Secunderabad,
500100, (2026-2027)

ABSTRACT

- RCC beams are structural members used to transfer loads safely to columns and supports.
- This project presents RCC beam reinforcement detailing and beam layouts prepared for AutoCAD drafting.
- The actual values and dimensions shown in the provided drawings are considered in this presentation.
- The project includes main reinforcement, extra bars, stirrups, clear cover and PB10/PB11 beam layouts.

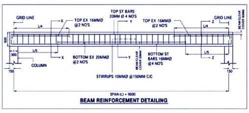
OBJECTIVES OF THE PROJECT

- To understand RCC beam reinforcement detailing.
- To study beam span, dimensions and reinforcement arrangement.
- To identify top bars, bottom bars, extra bars and stirrups.
- To prepare professional structural detailing using AutoCAD.

DESIGN AND REINFORCEMENT DATA

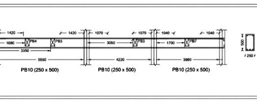
Beam span (L)	9000 mm
Beam depth	500 mm
Beam width shown	300 mm
Beam clear cover	30 mm
Top straight bars	4 Nos. Ø20 mm
Top extra bars	2 Nos. Ø16 mm
Bottom straight bars	4 Nos. Ø16 mm
Bottom extra bars	2 Nos. Ø20 mm
Stirrups	Ø10 mm @ 150 mm C/C

BEAM REINFORCEMENT DETAILING


GRID LINE
TOP EX 16MMØ @2 NO'S
TOP ST BARS 20MM Ø 4 NO'S
TOP EX 16MMØ @2 NO'S
GRID LINE
BEAM CLEAR COVER 30MM
X
X
Y
Y
Z
Z
L/4
L/4
L/5
L/5
500
300
COLUMN
BOTTOM EX 20MMØ @2 NO'S
BEAM CLEAR COVER 30MM
BOTTOM ST BARS 16MMØ @4 NO'S
150
150
STIRRUPS 10MMØ @150MM C/C
SPAN (L) = 9000

BEAM REINFORCEMENT DETAILING

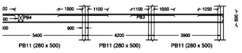
RCC BEAM LAYOUT – PB10 (250 × 500)


1420
1420
1070
1070
1040
1040
1680
PB4
PB3
3050
PB3
1700
PB7
3350
5550
4220
3980
500
250
PB10 (250 x 500)
PB10 (250 x 500)
PB10 (250 x 500)

EXPLANATION OF REINFORCEMENT DETAIL

- Total beam span shown in the drawing: L = 9000 mm.
- Top extra reinforcement is provided over support zones as indicated by L/4.
- Bottom extra reinforcement is provided as indicated by L/5.
- Top and bottom straight bars continue along the beam length.
- Stirrups of Ø10 mm are provided at 150 mm C/C.
- Beam clear cover is shown as 30 mm.

RCC BEAM LAYOUT – PB11 (280 × 500)


00
1500
1100
1100
1050
1050
00
PB4
PB3
00
900
5400
4200
3900
500
PB11 (280 x 500)
PB11 (280 x 500)
PB11 (280 x 500)

AUTOCAD DRAWING PROCEDURE

- Set units to millimetres.
- Create separate layers for beams, reinforcement, dimensions and text.
- Draw columns and beam layout according to the given dimensions.
- Add top, bottom and extra reinforcement bars.
- Provide stirrups at the specified spacing.
- Add dimensions, reinforcement notes and labels.
- Check the final drawing before plotting.

CONCLUSION

- This presentation uses the actual RCC beam reinforcement and layout values shown in the provided diagrams.
- The final project includes detailed reinforcement drawing and PB10/PB11 beam layouts.
- AutoCAD helps prepare clear, accurate and professional structural detailing drawings.

THANK YOU

CONCLUSION

- This presentation uses the actual RCC beam reinforcement and layout values shown in the provided diagrams.
- The final project includes detailed reinforcement drawing and PB10/PB11 beam layouts.
- AutoCAD helps prepare clear, accurate and professional structural detailing drawings.

THANK YOU